

Work, Energy and Power

A second way to solve mechanics problems: track energy instead of forces, and many hard problems become easy.

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Newton's laws can, in principle, solve any problem in mechanics. In practice, when forces vary with position or the path is complicated (a roller coaster, a pendulum, a spring), integrating $\vec{F} = m\vec{a}$ is hard. **Energy methods** replace a vector equation in time with a scalar bookkeeping rule that relates states. “How fast is it moving when it gets *there*?” can often be answered in one line. Energy is also the currency that links mechanics to heat, electricity, chemistry and biology. It is how engineers size motors and power plants, and how physiologists measure what the body does.

YOU WILL BE ABLE TO

- Calculate the work done by constant and variable forces, including as an area under a force–displacement graph.
- Derive and apply the work–energy theorem.
- Distinguish conservative from non-conservative forces, and define potential energy for each conservative force.
- Obtain force from potential energy ($F = -dU/dx$), and read energy diagrams: turning points, and stable and unstable equilibria.
- Apply conservation of mechanical energy, and the generalised energy equation with friction.
- Calculate power and efficiency in mechanical, biological and engineering systems.

THINK FIRST

A ball is thrown from the top of a cliff at the same speed three times: once straight up, once horizontally, and once straight down. Ignoring air resistance, which throw hits the sea at the highest speed?

All three hit the sea at the **same speed**. Each starts with the same kinetic energy and loses the same gravitational potential energy on the way down to sea level. Gravity does work mgh regardless of the path. The direction of the throw changes the *time* taken and the *direction* of impact, but not the speed. With Newton's laws you would need three separate projectile calculations to see this. With energy it takes one line.

§5.1

Work done by a force

Constant force

A constant force $\vec{F}$ acting on a body that undergoes a displacement $\vec{d}$ does **work**

$$W = \vec{F} \cdot \vec{d} = Fd \cos \theta,$$

where θ is the angle between the force and the displacement. Work is a scalar, measured in **joules**: $1 \text{ J} = 1 \text{ N m}$.

- $0 \leq \theta < 90^\circ$: positive work. The force helps the motion.
- $\theta = 90^\circ$: zero work. Examples are the normal force on a block sliding along a level floor, and the tension in the string of a body moving in a circle.
- $90^\circ < \theta \leq 180^\circ$: negative work. The force opposes the motion, as kinetic friction does.

When several forces act, the **net work** is the sum of the work done by each force, which equals the work done by the net force.

WORKED EXAMPLE 5.1

Work done by each force

A 20 kg crate is dragged 10 m across a floor by a rope pulling with 100 N at 30° above the horizontal. The coefficient of kinetic friction is 0.30. Find the work done by each force and

the net work. If the crate starts from rest, how fast is it moving at the end?

Solution

Normal force and friction. The rope supports part of the crate's weight:

$$N = mg - T \sin 30^\circ = 196 - 50 = 146 \text{ N}, \quad f_k = \mu_k N = 0.30(146) = 43.8 \text{ N}.$$

Work by each force over $d = 10 \text{ m}$:

FORCE	ANGLE TO $\vec{d}$	WORK
Tension, 100 N	30°	$100(10) \cos 30^\circ = +866 \text{ J}$
Friction, 43.8 N	180°	-438 J
Weight, 196 N	90°	0
Normal force, 146 N	90°	0
Net		$+428 \text{ J}$

Final speed. By the work-energy theorem (next section), $W_{\text{net}} = \frac{1}{2}mv^2 - 0$, so

$$v = \sqrt{\frac{2(428)}{20}} = 6.54 \text{ m/s}.$$

Variable force

If the force varies along the path, divide the path into small steps $d\vec{r}$ and add up the work done in each step:

$$W = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r}. \quad \text{In one dimension: } W = \int_{x_1}^{x_2} F(x) dx.$$

Geometrically, in one dimension W is the area under the F - x graph.

Work done stretching a spring. An ideal spring exerts $F_s = -kx$ when its extension is x . To stretch it slowly, you must apply $F = +kx$. The work you do in stretching it from x_1 to x_2 is

$$W = \int_{x_1}^{x_2} kx dx = \frac{1}{2}kx_2^2 - \frac{1}{2}kx_1^2.$$

This is the area of a trapezium under the straight line $F = kx$. The spring itself does the negative of this amount of work.

WORKED EXAMPLE 5.2

Stretching a spring further

A spring with $k = 400$ N/m is already stretched by 0.10 m. How much work is needed to stretch it to 0.30 m? Why is this more than twice the work needed to stretch it from 0 to 0.10 m and then again by a further 0.10 m?

Solution

$$W = \frac{1}{2}k(x_2^2 - x_1^2) = \frac{1}{2}(400)(0.09 - 0.01) = 200(0.08) = 16 \text{ J}.$$

The first 0.10 m needs only $\frac{1}{2}(400)(0.01) = 2$ J. The second 0.10 m needs $\frac{1}{2}(400)(0.04 - 0.01) = 6$ J, and the third needs $\frac{1}{2}(400)(0.09 - 0.04) = 10$ J.

Evaluate. Each equal extension costs more than the last, because the spring force grows linearly with x . The total work grows as x^2 , which is why the energy stored in a spring (and in a stretched tendon) rises steeply with deformation.

§5.2

The work–energy theorem

Define the **kinetic energy** of a particle of mass m moving at speed v :

$$K = \frac{1}{2}mv^2.$$

Derivation. Take the motion along x , where the net force is $F_{\text{net}} = ma = m v \frac{dv}{dx}$ (using $a = v \frac{dv}{dx}$ from Chapter 1). Then

$$W_{\text{net}} = \int_{x_1}^{x_2} F_{\text{net}} dx = \int_{x_1}^{x_2} m v \frac{dv}{dx} dx = \int_{v_1}^{v_2} m v dv = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2.$$

The same argument works in three dimensions with $\vec{F} \cdot d\vec{r}$.

KEY IDEA

Work–energy theorem. The net work done on a particle equals the change in its kinetic energy:

$$W_{\text{net}} = \Delta K = K_f - K_i.$$

This is Newton’s second law integrated over displacement. It holds for any force, constant or not, conservative or not.

WORKED EXAMPLE 5.3**Skid marks**

A 1200 kg car skids to a stop with its wheels locked, leaving skid marks 45.6 m long. The coefficient of kinetic friction between the tyres and the road is 0.70. How fast was the car going when it began to skid?

Solution

Only friction does work: the weight and the normal force are perpendicular to the motion.

$$W_f = -\mu_k mgd = \Delta K = 0 - \frac{1}{2}mv_0^2 \Rightarrow v_0 = \sqrt{2\mu_k g d}.$$

$$v_0 = \sqrt{2(0.70)(9.8)(45.6)} = \sqrt{625.6} = 25.0 \text{ m/s (90 km/h)}.$$

Evaluate. The mass cancels, so accident investigators can estimate speed from skid length without knowing the car’s mass. Since $d \propto v_0^2$, doubling the speed quadruples the skid length.

§5.3

Conservative forces and potential energy

A force is **conservative** if the work it does on a body moving between two points depends only on the end points, not on the path taken. Equivalently, the work it does around any **closed** path is zero. Gravity, the spring force and the electrostatic force are conservative. Kinetic friction and air drag are **non-conservative**: they always take energy away, and a longer path means more work done against them.

For each conservative force we can define a **potential energy** U , energy stored in the configuration of the system, such that

$$W_{\text{cons}} = -\Delta U = U_i - U_f.$$

Only *changes* in U are physical, so we may choose where $U = 0$ to suit the problem.

FORCE	POTENTIAL ENERGY	TYPICAL ZERO
Uniform gravity, mg downward	$U_g = mgy$	any convenient height
Spring, $-kx$	$U_s = \frac{1}{2}kx^2$	natural length
Newtonian gravity, $-GMm/r^2$	$U = -GMm/r$	$r \rightarrow \infty$ (Chapter 8)
Coulomb force	$U = kq_1q_2/r$	$r \rightarrow \infty$ (Chapter 15)

Force from potential energy

Reversing $\Delta U = -\int F dx$ gives

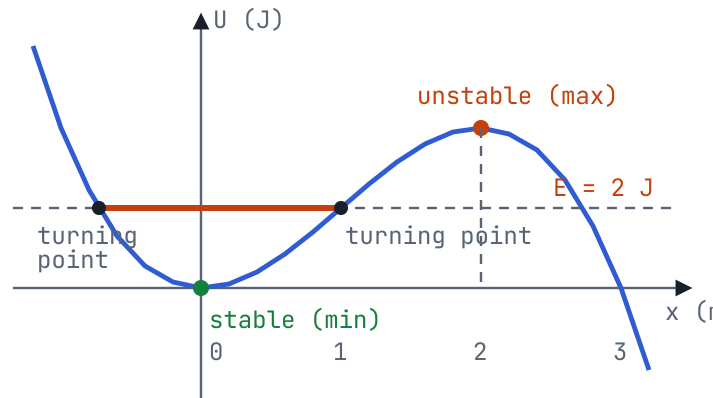
$$F_x = -\frac{dU}{dx} \quad (\text{in 3D: } \vec{F} = -\nabla U).$$

The force points **downhill** on the potential-energy curve, towards lower U .

Energy diagrams and equilibrium

Plot $U(x)$ and draw a horizontal line at the total mechanical energy E . Since $K = E - U \geq 0$, the particle can only be where $U(x) \leq E$. Where the line meets the curve, $K = 0$: these are the **turning points**.

- **Equilibrium** occurs where $dU/dx = 0$, so that $F = 0$.
- **Stable** equilibrium is at a *minimum* of U ($d^2U/dx^2 > 0$). A small displacement produces a restoring force.
- **Unstable** equilibrium is at a *maximum* of U ($d^2U/dx^2 < 0$). A small displacement grows.
- **Neutral** equilibrium is where U is flat.



Energy diagram for $U = 3x^2 - x^3$. With $E = 2$ J, the particle is trapped between the turning points at $x \approx -0.73$ m and $x = 1$ m, oscillating about the stable minimum at $x = 0$.

WORKED EXAMPLE 5.4

Reading an energy diagram

A particle moving along x has potential energy $U(x) = 3x^2 - x^3$ (U in J, x in m), as in the figure. (a) Find the force $F(x)$. (b) Locate the equilibrium points and classify them. (c) The particle has total energy $E = 2$ J and starts near $x = 0$. Find its turning points. (d) What is the minimum energy it needs to escape to large positive x ?

Solution

(a) $F = -\frac{dU}{dx} = -6x + 3x^2$.

(b) $F = 0$ where $3x(x - 2) = 0$, so $x = 0$ and $x = 2$ m. The second derivative is $U''(x) = 6 - 6x$:

- $U''(0) = 6 > 0$, so $x = 0$ is a minimum: **stable**.
- $U''(2) = -6 < 0$, so $x = 2$ is a maximum: **unstable**, with $U(2) = 12 - 8 = 4$ J.

(c) Turning points are where $U = E$:

$$3x^2 - x^3 = 2 \Rightarrow x^3 - 3x^2 + 2 = 0.$$

$x = 1$ is a root by inspection, so factorise: $(x - 1)(x^2 - 2x - 2) = 0$, giving $x = 1$ or $x = 1 \pm \sqrt{3}$. Starting near the origin, the particle is confined between

$x = 1 - \sqrt{3} = -0.73$ m and $x = 1$ m. The third root, 2.73 m, lies beyond the barrier and cannot be reached.

(d) To pass over the barrier at $x = 2$, the particle needs $E > U(2) = 4 \text{ J}$.

Evaluate. This is the same picture as an atom bound in a molecule, or a ball in a dip on a hillside. Below the barrier the particle is trapped and oscillates. Above it, the particle escapes. Quantum mechanics later adds the surprise that a particle can *tunnel* through a barrier it classically cannot cross (Chapter 24).

§5.4

Conservation of mechanical energy

If only conservative forces do work, then $W_{\text{net}} = W_{\text{cons}} = -\Delta U$. Combining this with the work-energy theorem, $W_{\text{net}} = \Delta K$, gives $\Delta K + \Delta U = 0$.

KEY IDEA

Conservation of mechanical energy. If only conservative forces do work, the mechanical energy $E = K + U$ stays constant:

$$K_i + U_i = K_f + U_f.$$

Forces that do no work, such as the normal force on a fixed surface or the tension in a pendulum string, may act without spoiling this.

WORKED EXAMPLE 5.5

The minimum height for a loop-the-loop

A small cart starts from rest at height h on a frictionless track and enters a vertical loop of radius R at ground level. Find the minimum h for the cart to stay on the track at the top of the loop. Then find the normal force at the bottom of the loop in this case.

Solution

Condition at the top. From Chapter 4, the cart stays on the track only if $v_{\text{top}}^2 \geq gR$.

Energy, start to top. The top of the loop is at height $2R$:

$$mgh = mg(2R) + \frac{1}{2}mv_{\text{top}}^2 \geq 2mgR + \frac{1}{2}mgR \Rightarrow h_{\text{min}} = \frac{5}{2}R.$$

At the bottom, with $h = \frac{5}{2}R$: $v_b^2 = 2gh = 5gR$. The radial equation gives

$$N - mg = \frac{mv_b^2}{R} = 5mg \Rightarrow N = 6mg.$$

Evaluate. The extra $\frac{1}{2}R$ above the top of the loop provides the kinetic energy needed for the centripetal requirement at the top. The 6g load at the bottom is why real roller coasters use teardrop-shaped (clothoid) loops: they are tighter at the top and gentler at the bottom, keeping riders' loads near 3–4 g.

WORKED EXAMPLE 5.6

A spring launcher

A spring with $k = 800$ N/m is compressed by 0.15 m and used to launch a 50 g ball vertically upward. How high does the ball rise above its launch position (the compressed position)? Ignore air resistance.

Solution

Take the launch position as $y = 0$. The spring energy becomes gravitational energy at the top, where the ball is momentarily at rest:

$$\frac{1}{2}kx^2 = mgh \Rightarrow h = \frac{kx^2}{2mg} = \frac{800(0.0225)}{2(0.050)(9.8)} = \frac{18}{0.98} = 18.4 \text{ m}.$$

Evaluate. The 9 J of stored energy is modest, but the ball is light. The maximum speed occurs not at the moment the ball leaves the spring but slightly before, where the spring force equals the weight, $kx = mg$, that is, $x = 0.6$ mm before release. For such a stiff spring the difference is negligible.

§5.5

When friction acts: the general energy equation

If non-conservative forces do work W_{nc} , then

$$K_i + U_i + W_{nc} = K_f + U_f, \quad \text{or} \quad \Delta E_{\text{mech}} = W_{nc}.$$

Kinetic friction makes $W_{nc} = -f_k d < 0$. Mechanical energy is converted into **thermal energy**: the surfaces warm up. Total energy (mechanical + thermal + chemical + ...) is always conserved. This is the first law of thermodynamics, which we meet in Chapter 12.

WORKED EXAMPLE 5.7

Energy lost on a playground slide

A 25 kg child starts from rest at the top of a slide 3.0 m high and reaches the bottom at 6.0 m/s. How much energy was converted to thermal energy? If the slide is 5.0 m long, what is the average friction force?

Solution

$$W_{\text{nc}} = \Delta K + \Delta U = \frac{1}{2}(25)(6.0)^2 - 25(9.8)(3.0) = 450 - 735 = -285 \text{ J.}$$

So 285 J became thermal energy, about 39% of the initial potential energy.

Average friction force: $f = 285/5.0 = 57 \text{ N.}$

§5.6

Power

Power is the rate of doing work, or of transferring energy:

$$P = \frac{dW}{dt}, \quad \text{average } \bar{P} = \frac{W}{\Delta t}.$$

The SI unit is the watt ($1 \text{ W} = 1 \text{ J/s}$). An older unit is 1 horsepower = **746 W**. For a force acting on a moving body, $dW = \vec{F} \cdot d\vec{r}$, so

$$P = \vec{F} \cdot \vec{v}.$$

A machine's efficiency is

$$\eta = \frac{\text{useful power out}}{\text{power in}}.$$

WORKED EXAMPLE 5.8

Why fuel use rises steeply with speed

A 1500 kg car has a drag coefficient of 0.30 and a frontal area of **2.2 m²**, in air of density **1.2 kg/m³**. Its rolling resistance is 1% of its weight. Find the power needed to cruise on a level road at (a) 30 m/s (108 km/h) and (b) 40 m/s (144 km/h).

Solution

At constant speed the engine force equals the total resistance, so $P = Fv$.

- Rolling resistance: $0.010(1500)(9.8) = 147$ N, the same at all speeds.
- Air drag: $D = \frac{1}{2}C\rho Av^2 = \frac{1}{2}(0.30)(1.2)(2.2)v^2 = 0.396v^2$.

(a) At 30 m/s: $D = 356$ N, so $F = 503$ N and $P = 503(30) = 15.1$ kW.

(b) At 40 m/s: $D = 634$ N, so $F = 781$ N and $P = 781(40) = 31.2$ kW.

Evaluate. A 33% increase in speed **doubles** the power. At high speed, drag dominates, and drag power grows as v^3 . Energy *per kilometre* is force $\times$ distance, which grows as v^2 . That is why fuel economy falls steeply above about 90 km/h, and why streamlining matters so much for electric vehicles' range.

WORKED EXAMPLE 5.9

The power of climbing stairs

A 70 kg student runs up five flights of stairs (15 m of height) in 30 s. Find (a) the mechanical power output and (b) the metabolic power and food energy used, if muscles are 25% efficient.

Solution

(a) Mechanical power:

$$P = \frac{mgh}{t} = \frac{70(9.8)(15)}{30} = \frac{10\,290 \text{ J}}{30 \text{ s}} = 343 \text{ W}.$$

(b) Metabolic power: $343/0.25 = 1372$ W. Over 30 s this uses

$$1372 \times 30 = 41 \text{ kJ} \approx 10 \text{ kcal (food Calories)}, \quad 1 \text{ kcal} = 4.184 \text{ kJ}.$$

Evaluate. The other 75% (about 1 kW) is released as heat, which is why you warm up quickly when exercising. A chocolate bar (about 250 kcal) would fuel some 25 such climbs. This kind of calculation underlies exercise physiology and the design of cardiac stress tests.

IN PRACTICE

Energy in engineering and the body.

- **Regenerative braking.** Hybrid and electric cars run their motors as generators when braking, turning kinetic energy back into stored chemical energy instead of brake-pad heat. This recovers about 60–70% of the braking energy.
- **Pumped hydro.** A pumped-storage station pumps water uphill when electricity is cheap and runs it back down through turbines when demand peaks. It is the world's largest form of grid energy storage, with a round-trip efficiency of about 75–80%.
- **Elastic tendons.** In running, the Achilles tendon stores elastic potential energy as the foot lands and returns most of it at push-off, like a spring. This substantially reduces the metabolic cost of running.
- **The heart.** The heart's mechanical power output is only about 1 W (Problem P5.10), yet it must run continuously for a lifetime.

CHAPTER SUMMARY

- Work: $W = \vec{F} \cdot \vec{d}$ for a constant force, and $W = \int \vec{F} \cdot d\vec{r}$ (the area under F - x) in general. The work needed to stretch a spring is $\frac{1}{2}k(x_2^2 - x_1^2)$.
- Kinetic energy $K = \frac{1}{2}mv^2$. Work-energy theorem: $W_{\text{net}} = \Delta K$.
- Conservative forces do path-independent work, $W = -\Delta U$. Examples: $U_g = mgy$, $U_s = \frac{1}{2}kx^2$. Force from potential: $F = -dU/dx$.
- Energy diagrams: the motion is confined to $U \leq E$, with turning points where $U = E$. Minima of U are stable and maxima unstable.
- Only conservative forces doing work: $K + U$ is constant. With friction: $\Delta E_{\text{mech}} = W_{\text{nc}}$ (lost to heat).
- Power $P = dW/dt = \vec{F} \cdot \vec{v}$. Efficiency $\eta = P_{\text{out}}/P_{\text{in}}$. Drag power grows as v^3 .

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P5.1

Can kinetic friction ever do *positive* work on a body? Can static friction? Give an example of each, or explain why not.

P5.2

Does the normal force ever do work? Give an example in which it does positive work and one in which it does none.

P5.3

A satellite moves in a circular orbit at constant speed. How much work does gravity do on it during one complete orbit? During a quarter of an orbit? Explain.

P5.4

Two identical blocks slide from rest down two frictionless ramps of the same height. One ramp is steep and short, the other gentle and long. Compare their final speeds and the times taken.

LEVEL B – STANDARD PROBLEMS

P5.5

A force $F(x) = (3x^2 - 2x)$ N acts on a particle moving along the x -axis. Find the work it does as the particle moves from $x = 1$ m to $x = 3$ m.

P5.6

A 2.0 kg block slides at 4.0 m/s across a frictionless floor and runs into a horizontal spring with $k = 200$ N/m. (a) Find the maximum compression of the spring. (b) Repeat part (a) if the floor under the spring has $\mu_k = 0.20$.

P5.7

A 0.50 kg pendulum bob on a 2.0 m string is released from rest with the string at 60° to the vertical. Find its speed at the lowest point and the tension in the string there.

P5.8

A 1500 kg car climbs a road with a 5% grade (it rises 5 m for every 100 m travelled along the road) at a constant 20 m/s. Air and rolling resistance total 600 N. Find the power the engine must deliver to the wheels.

P5.9

A 70 kg skier starts from rest and descends a slope with a vertical drop of 40 m, reaching the bottom at 22 m/s. How much mechanical energy was lost to friction and air resistance? What fraction of the initial potential energy is this?

P5.10

Cardiac work. The left ventricle pumps about 5.0 L of blood per minute into the aorta against an average pressure of about 13 kPa (100 mmHg). The work done in pushing a volume ΔV against pressure p is $p \Delta V$. (a) Find the mechanical power output of the left ventricle. (b) How much work does it do in a day? (c) If cardiac muscle is about 15% efficient, what metabolic power does the left ventricle need? (d) During heavy exercise, cardiac output can rise to 25 L/min and mean pressure to 16 kPa. By what factor does the ventricle's mechanical power increase?

P5.11

A hydroelectric dam passes $500 \text{ m}^3/\text{s}$ of water through turbines 100 m below the reservoir surface. If the turbines and generators together are 90% efficient, what is the electrical power output?

P5.12

A ball dropped from 10 m rebounds to a height of 6.4 m. (a) What fraction of its mechanical energy is lost in the bounce? (b) Find its speeds just before and just after the bounce. (c) Assuming the same fraction is lost at every bounce, how high does it rise after the fourth bounce?

LEVEL C – CHALLENGE PROBLEMS

P5.13

Bungee jumping. A 70 kg jumper uses a cord with an unstretched length of 20 m and a spring constant of 250 N/m. (a) How far does the jumper fall before coming momentarily to rest? (b) What is the maximum deceleration, in multiples of g ? (c) At what point in the fall is the jumper's speed greatest, and what is that speed?

P5.14

A small block slides from rest down a frictionless track from height $h = 3R$ and goes round a vertical loop of radius R . Find the normal force on the block (in terms of mg) at the bottom of the loop, at the point level with the centre of the loop, and at the top.

P5.15

A small particle slides from rest at the top of a smooth, fixed hemisphere of radius R , given only a tiny nudge. At what angle from the vertical, and at what height above the base, does it leave the surface? What is its speed at that moment?

P5.16

The interaction between two atoms in a diatomic molecule can be modelled by the potential energy $U(r) = \frac{A}{r^2} - \frac{B}{r}$, where A and B are positive constants and r is the separation. (a) Find the equilibrium separation r_0 and show that it is stable. (b) Find the energy needed to separate the atoms completely, starting from rest at r_0 (the dissociation energy). (c) For small oscillations about r_0 , the potential can be approximated as $\frac{1}{2}k_{\text{eff}}(r - r_0)^2$. Find k_{eff} .

P5.17

A uniform chain of length L lies on a frictionless table with a length a hanging over the edge. It is released from rest. Find its speed at the moment the last link leaves the table.

P5.18

A car of mass m starts from rest on a level road, and its engine delivers a *constant power* P to the wheels. Ignore all resistive forces. (a) Find $v(t)$ and $x(t)$. (b) For $P = 60$ kW and $m = 1200$ kg, how long does it take to reach 27 m/s (about 100 km/h), and how far does it travel? (c) The model predicts an infinite acceleration at $t = 0$. What really limits the acceleration of a car starting from rest?

All worked examples and practice problems in these notes are original.
Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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